Week 1 Recitation

TA Suha Kim

ECON8711 (Micro 1A)

Aug 26, 2023

TA Introduction

Suha Kim

- Email: kim.8316@osu.edu
  - Feel free to reach out!
  - Please indicate in the subject field: *"[ECON8711]* $\sim$ *"*

    ➔ This would help me get to you quickly.

- Recitation: Fri 1:00 - 2:50PM, Enarson Classroom Building 311
  - 50 min talk - 10 min break - 50 min talk

- Office hour: By appointment, Zoom option available

Question 1, 2

"Lines in any direction"?

- Usefulness of a convex domain: " We don't know where to go if we can't go in *lines* (of any *direction*)"
- Usefulness of a concave objective function (when maximizing): "It allows us to look at *lines* around candidate maximizers to be actual maximizers"

Set convexity:
A set $C \in \mathbb{R}^N$ is convex when for all $x, y \in C$ and for all $\lambda \in [0, 1]$

$$\lambda x + (1 - \lambda) y \in C$$

- See λ as a gradient

Question 3

What are come potential issues that could occur when you do not have a concave objective and a convex constraint set? Do you think it is important that these sets are also topologically closed? Discuss your opinions.

- Qual-type question! (In my experience)
- Closedness
- Boundedness

Definitions

Question 3

Existence of a maximizer!

1. When non-concave objective

2. When non-convex constraint

Question 3

3. When not closed constraint

- Boundedness?

Function and Set convexity

- Function Convexity: Epigraph ver Part 2
  A function $f : \mathbb{R}^N \to \mathbb{R}$ is convex when the set

$$\{(x, r) \in \mathbb{R}^{N+1} | r \geq f(x)\}$$

  is convex.
- Function Quasi-Convextiy: Domain Version
  A function $f : \mathbb{R}^N \to \mathbb{R}$ is quasi-convex when the set

$$S_\alpha(f) = \{x \in \mathbb{R}^N | f(x) \leq \alpha\}$$

  is convex.

How to Optimize - Lagrangian

Problem:

$$\begin{aligned} &\min_{y \in \mathbb{R}^N} f(y) \\ \text{s.t. } &g_j(y) \leq 0 \text{ for } j = 1, \cdots, J \\ &a_k^T y = b_k \text{ for } k = 1, \cdots, K \end{aligned}$$

Lagrangian:

$$\mathcal{L}(y, \mu, \lambda) = f(y) + \sum_{j=1}^{J} \mu_j g_j(y) + \sum_{k=1}^{K} \lambda_k (a_k^T y - b_k)$$

where $\mu_j \geq 0, \lambda_k \in \mathbb{R}$

➔ Take first-order conditions and try to solve the problem

Question 4

Problem:

$$\min_{y\in\mathbb{R}^N} f(y)$$
$$\text{s.t. } g_j(y) \le 0 \text{ for } j = 1, \cdots, J$$
$$a_k^T y = b_k \text{ for } k = 1, \cdots, K$$

Lagrangian:

$$\mathcal{L}(y, \mu, \lambda) = f(y) + \sum_{j=1}^{J} \mu_j g_j(y) + \sum_{k=1}^{K} \lambda_k (a_k^T y - b_k)$$

where $\mu_j \ge 0, \lambda_k \in \mathbb{R}$

Why did we write down this separate problem? Try to explain your reasoning at a high level.

Question 5

Maybe you aren't sure about the why of a Lagrangian. Think briefly about the term outside of the $f(y)$ in the Lagrangian. Is $\mathcal{L}(y, \mu, \lambda)$ greater than, less than, or equal to $f(y)$ when the constraints are satisfied? Can you draw a picture of what these are like?

Question 6

In light of Q5, could you think of a name for the added part of the Lagrangian? What would you call it?
Suppose instead you had the function

$$\mathcal{G}(y) = f(y) - A\big(g_1(y), \cdots, g_J(y); a_1^T y - b_1, \cdots, a_K^T y - b_K\big)$$

where $A(0,0) = 0$, A is decreasing in its first J components and increasing away from zero for the final K terms. Do you think this is a useful way to write down an optimization problem? Describe your answer.

Question 6-1

In light of the previous questions, how can you interpret the Lagrangian?

Definition of closed set / Bounded set

- Closed: Complement of open sets
- Open: U is open if every point in U has a neighborhood contained in U.
- Bounded: Metric space (M, d) is bounded if there exists $r > 0$ such that for all $s, t \in S$, we have $d(s, t) < r$.

Back to the Slide